

CIS 6270

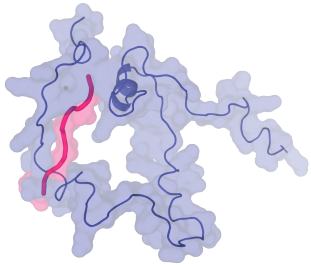
Module 5: Discrete Flow Matching

Lecture 5.1: Discrete Flow Matching with Jump Rates

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Let’s start by writing a rate as a velocity

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Could we choose a probability path first, then learn the token jumps that follow it?

Discrete Flow Matching

Discrete Flow Matching

Itai Gat[1] **Tal Remez**[1] **Neta Shaul**[2] **Felix Kreuk**[1] **Ricky T. Q. Chen**[1]

Gabriel Synnaeve[1] **Yossi Adi**[1] **Yaron Lipman**[1]

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Gat,..,Lipman. *NeurIPS* (2024)

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Time also flips from Lecture 4: here t = 0 is the source, or noise, and t = 1 is the data.

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The diagonal entry $u_t(x, x)$ equals negative the sum of the rates $u_t(y, x)$ from x to every other state y, which is the total rate of leaving x.

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A velocity specifies how probability moves. Let's now calculate how the probability of each state changes.

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We say that u_t generates the path p_t when this equation holds at every state y and every time t.

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In Lecture 2, density changed by the flux through the boundary of a region. Here it changes by the flux along the edges into and out of a state.

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The rate of change of $p_t(G)$ equals the probabilities 0.1, 0.2, and 0.4 of A, C, and T, each times the rate 1 toward G, which gives 0.7.

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The rates of change for A, C, G, and T are −0.1, −0.2, 0.7, and −0.4, and they sum to zero.

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Adding 0.1 times the rates of change to p_t gives 0.09, 0.18, 0.37, and 0.36, which still sum to one.

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Probability flows into G, and the total stays one.

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The velocity from the sequence x to the sequence y is the sum over positions i of the indicator that x and y agree everywhere except position i, times the per-position velocity $u^i_t(y^i, x)$ of changing position i to y^i.

Count the states of a DNA sequence

A DNA sequence of length N has 4^N states, so a rate for every pair of sequences is far too many numbers.

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Each position has only four rates, but those rates may depend on the whole current sequence x. The network computes those rates.

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$$\boxed{\Pr\left(X^i_{t+h} = y^i \mid X_t = x\right) = \delta_{x^i}(y^i) + h\, u^i_t(y^i, x)}$$

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From Lecture 4, a step stays valid only when h times the total rate of leaving x^i is at most one.

Multiply the per-position steps over all positions

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$$\prod_{i=1}^{N} \left[\delta_{x^i}(y^i) + h\, u_t^i(y^i, x)\right] = \delta_x(y) + h\, u_t(y, x) + O(h^2)$$

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So the parallel step agrees with the one-position-at-a-time process to first order in h, and generation can update all positions at once.

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As in Lecture 2, we build p_t by averaging simple paths, one for each pair of endpoints.

Take a coupling of the source and the data

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The independent coupling assigns each pair the source probability $p(x_0)$ times the data probability $q(x_1)$.

Common sources are the all-mask sequence (m, …, m) and uniformly random tokens. We return to choosing π when we study rectification.

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Given the pair, position i holds

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Positions are independent given the pair: $$p_t(x \mid x_0, x_1) = \prod_{i=1}^{N} p_t(x^i \mid x_0, x_1)$$

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The conditional probability of the whole sequence x is the product over positions i of the per-position probabilities.

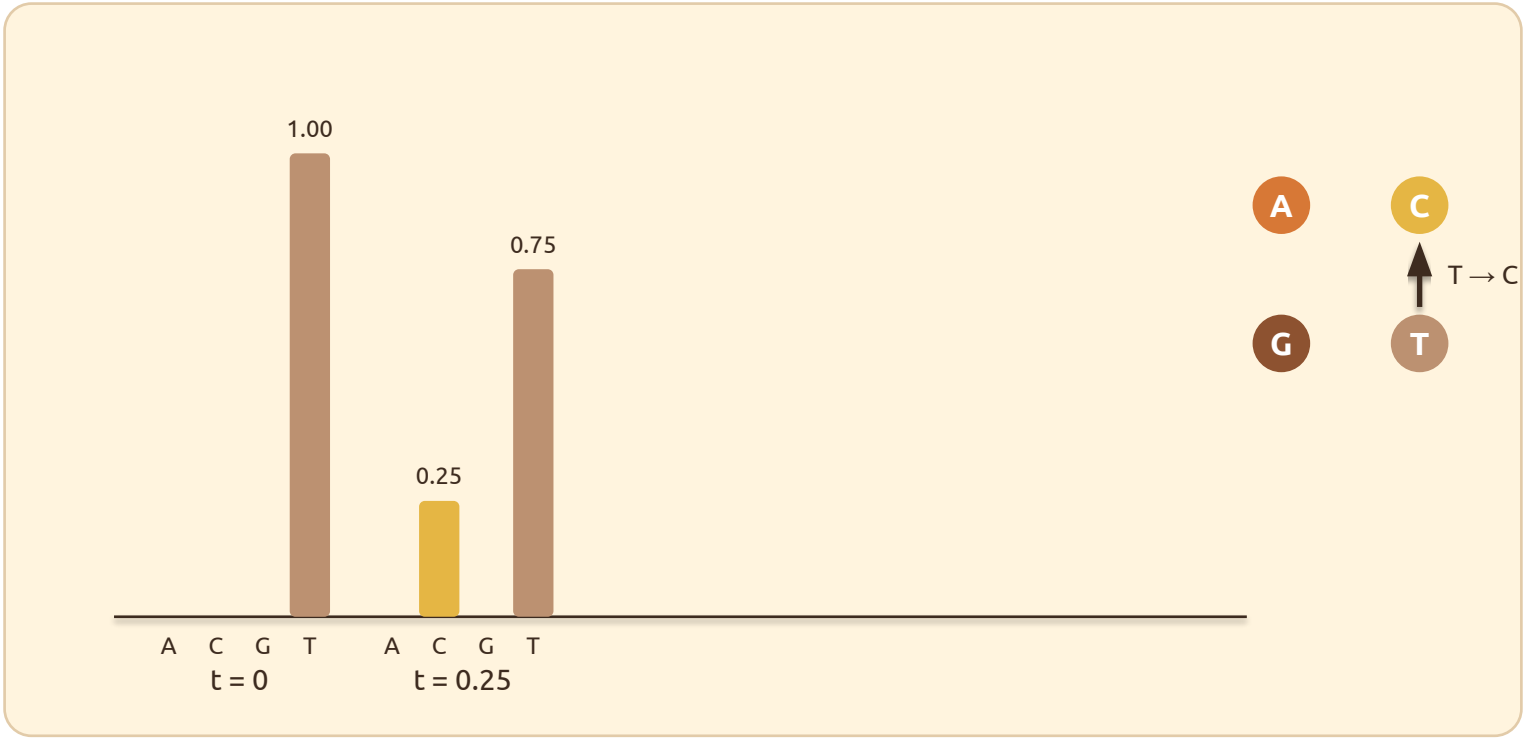
Let’s follow one position whose source token is T and data token is C



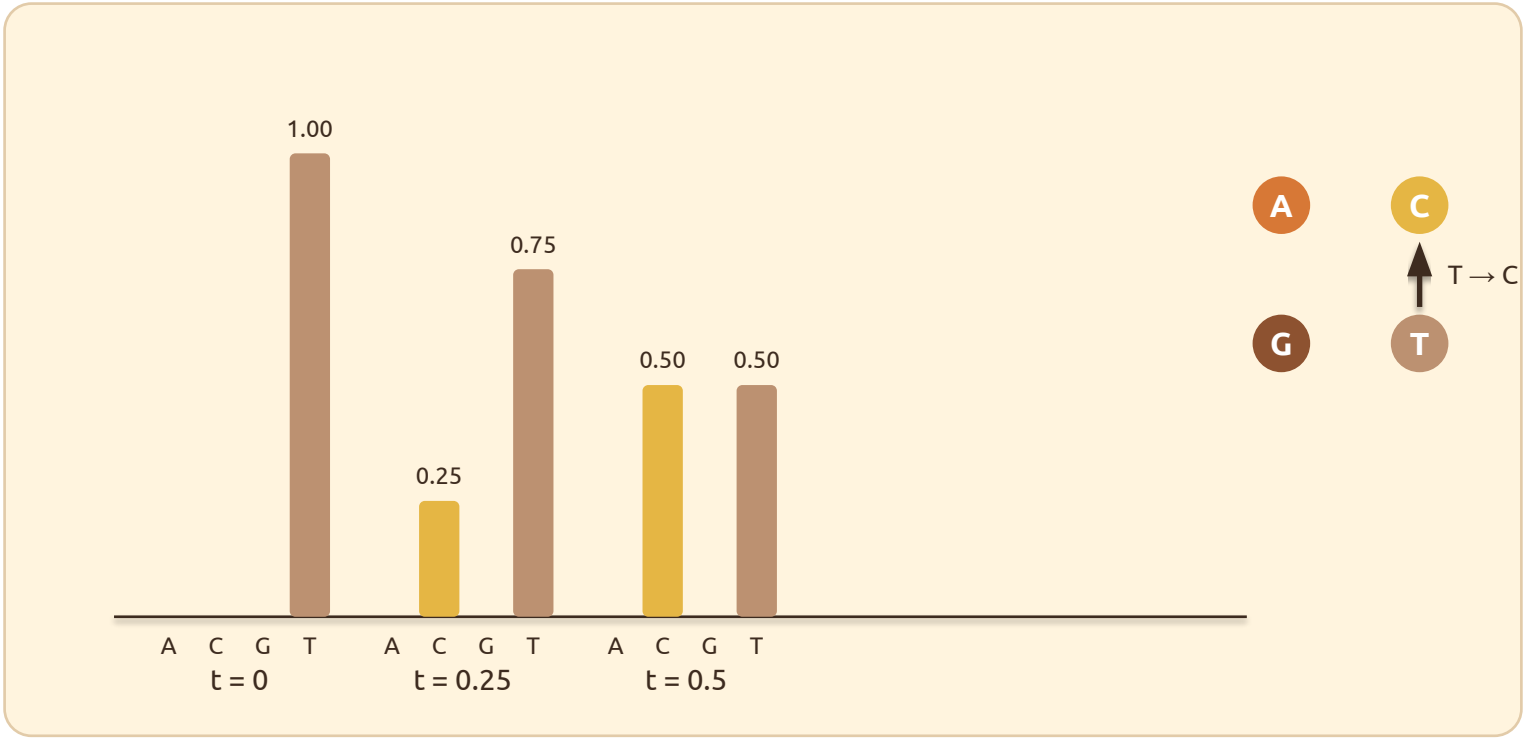
At t = 0, all of its probability sits on the source token T



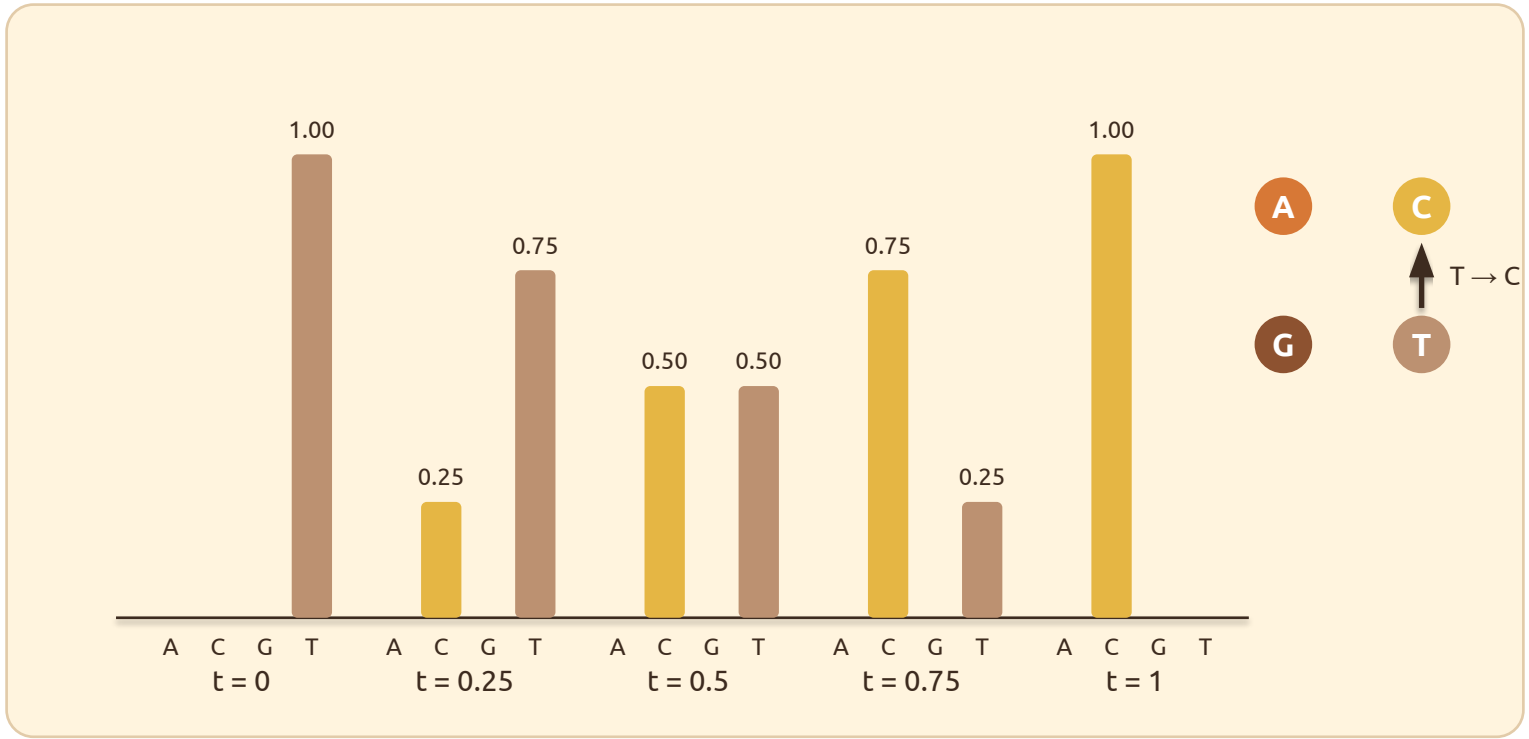
With $\kappa_t = t$, a quarter has moved to C by t = 0.25



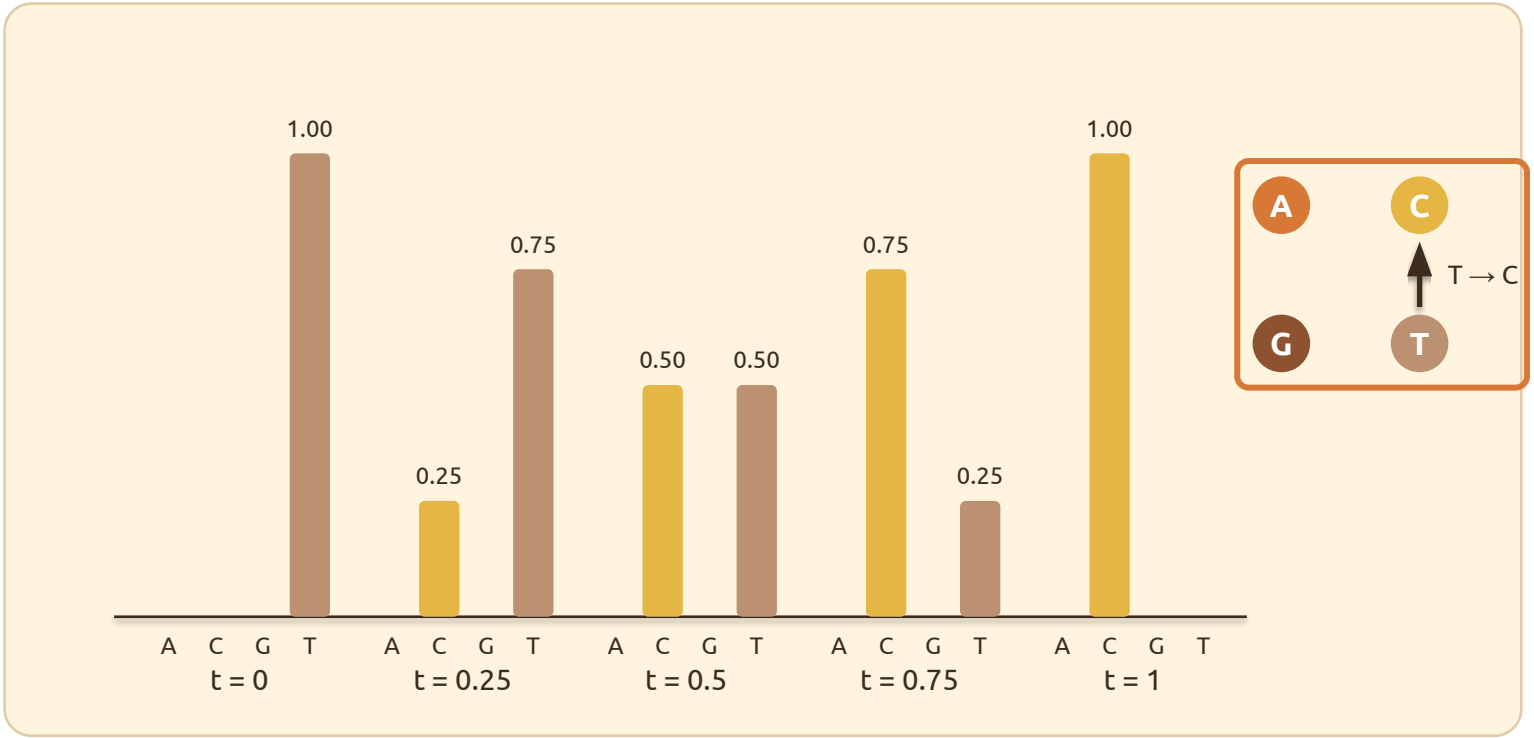
Halfway through, T and C are equally likely



At t = 1, the position shows its data token C



Probability moves along a single edge, from T to C



Now average the conditional paths over the coupling

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We never need to write $p_t(x)$ down. We only need to sample from it.

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ATGA is our noisy input, and ACGT is its training target

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Both are mixture paths, so one derivation covers them.

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Probability leaves the source token and arrives at the data token at rate $\dot{\kappa}_t$, so all of it moves along a single edge.

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It depends on the pair only through the data token x_1, and we will rely on this when we average.

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As fewer source tokens remain, each remaining one must jump faster to move the same probability per unit time.

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Substitute the path into the master equation. Only $z = x_0$ and $z = x_1$ carry probability:

$$\sum_z u_t(a, z \mid x_1)\, p_t(z \mid x_0, x_1) = (1 - \kappa_t)\, \frac{\dot{\kappa}_t}{1 - \kappa_t} \left[\delta_{x_1}(a) - \delta_{x_0}(a)\right] + \kappa_t \cdot 0$$

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The right side of the master equation equals one minus κ_t, the probability of the source token, times the velocity from x_0, plus κ_t, the probability of the data token, times the velocity from x_1, which is zero.

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$$= \dot{\kappa}_t \left[\delta_{x_1}(a) - \delta_{x_0}(a)\right] = \frac{d}{dt}\, p_t(a \mid x_0, x_1)$$

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The factors of one minus κ_t cancel, which leaves $\dot{\kappa}_t$ times the difference of indicators, and this is the rate of change of the conditional path.

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The velocity depends on x_1, but not on x_0.

Take a position still at its source token at time t

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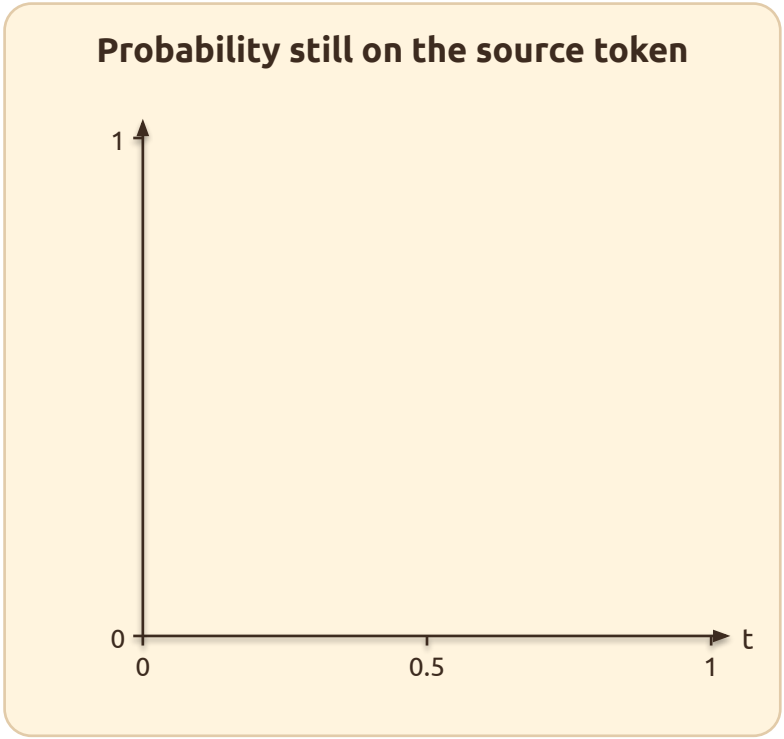
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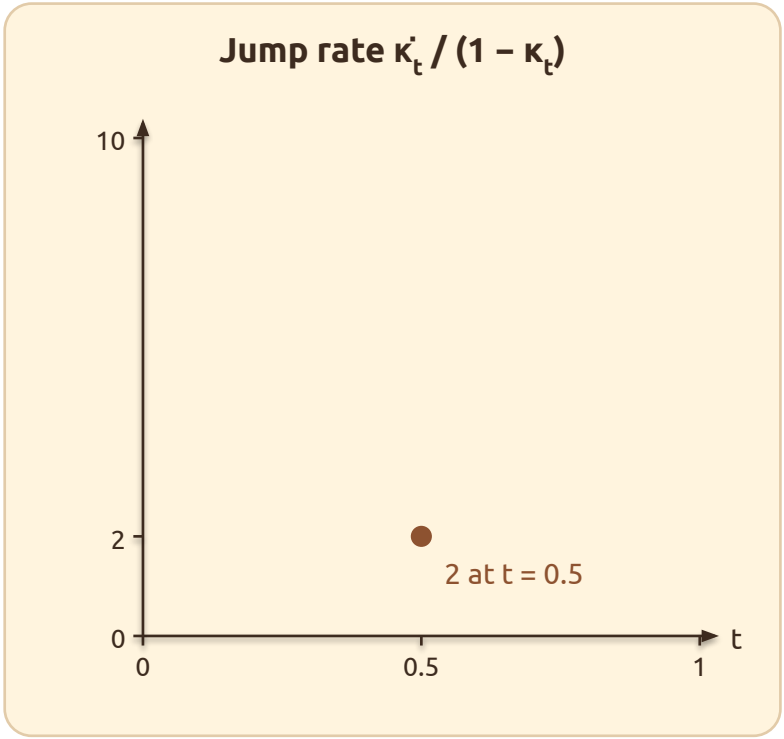
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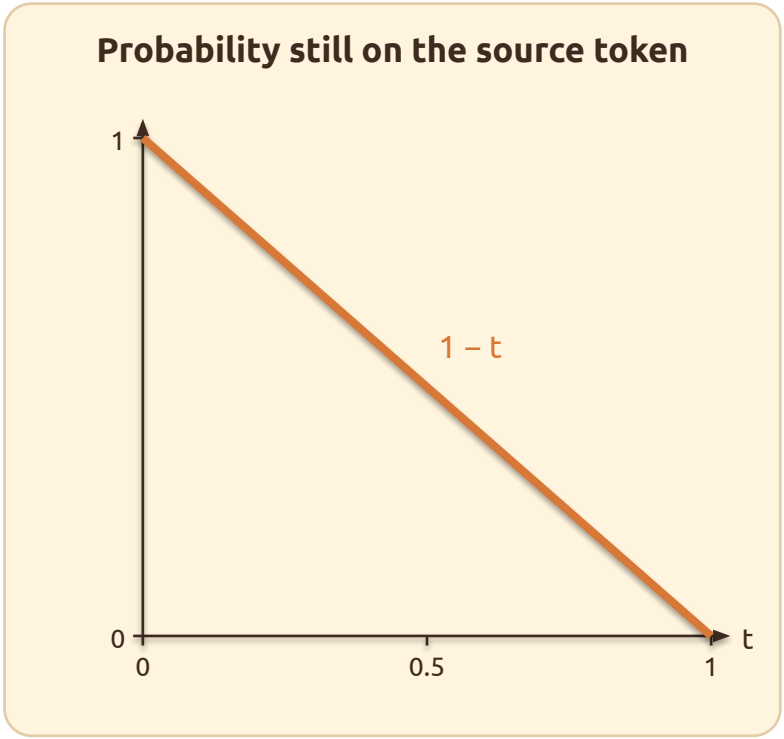
The rate diverges at t = 1, so every position reaches its data token by the end.

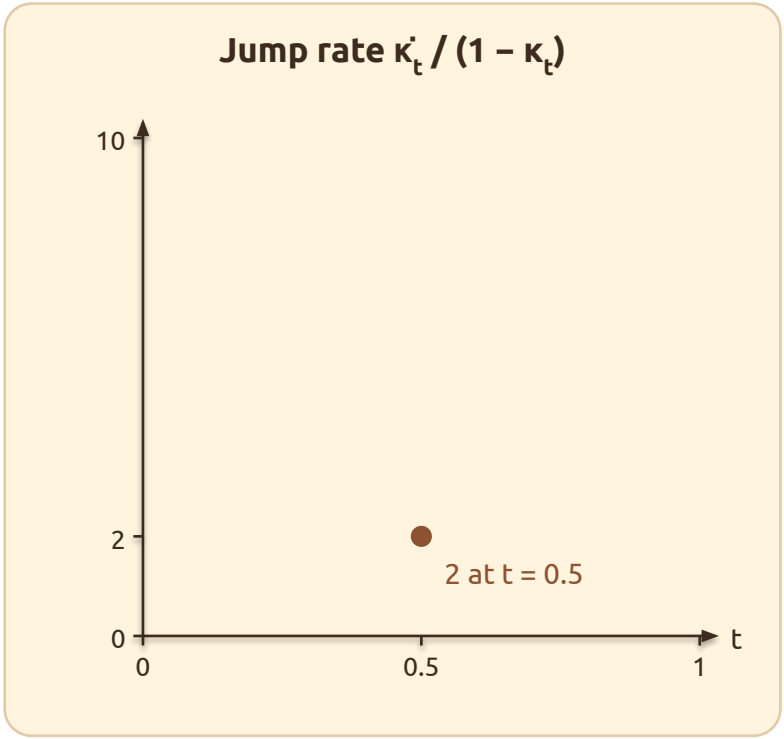
Now plot the survival probability and the jump rate for $\kappa_t = t$



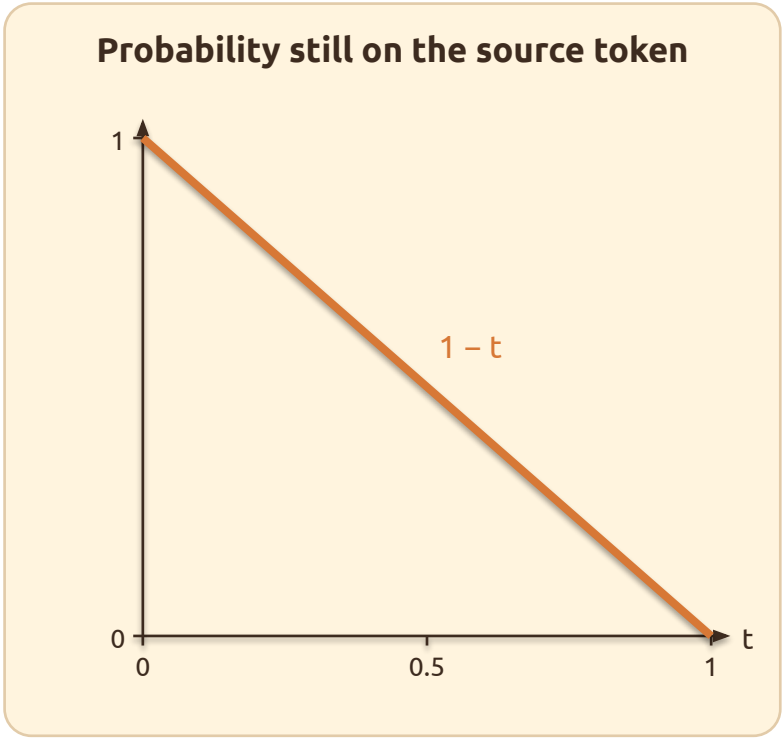


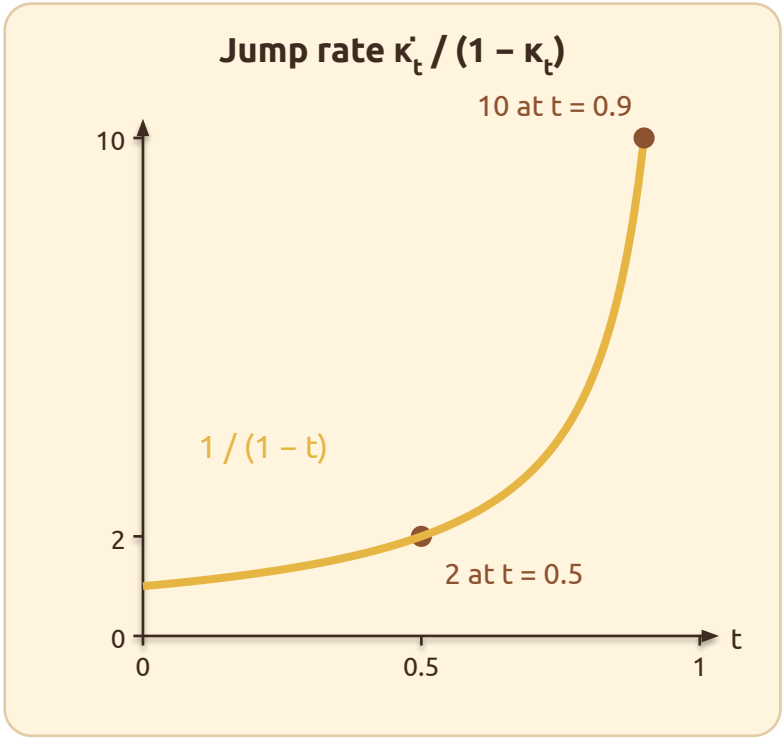
The survival probability $1 - \kappa_t$ falls in a straight line



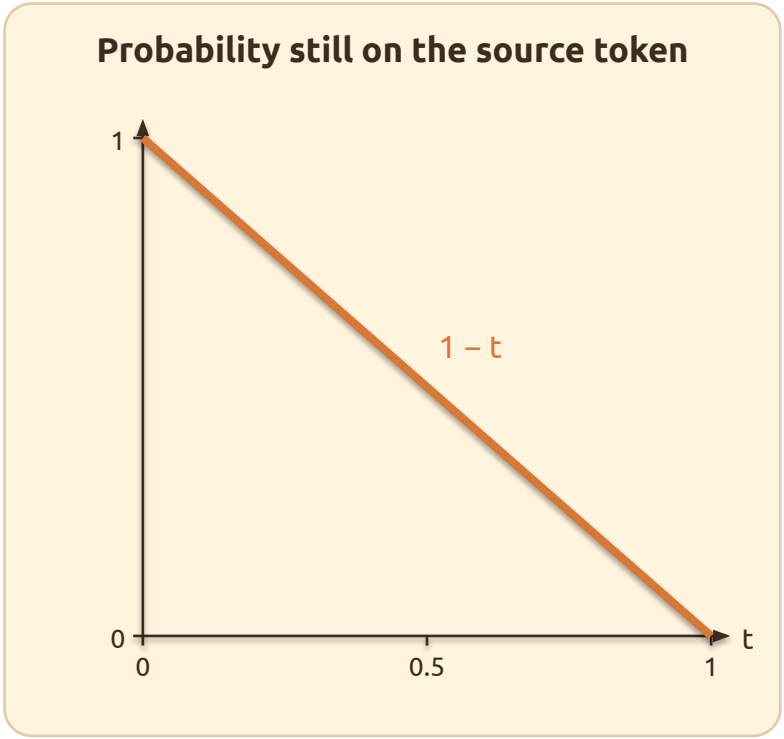


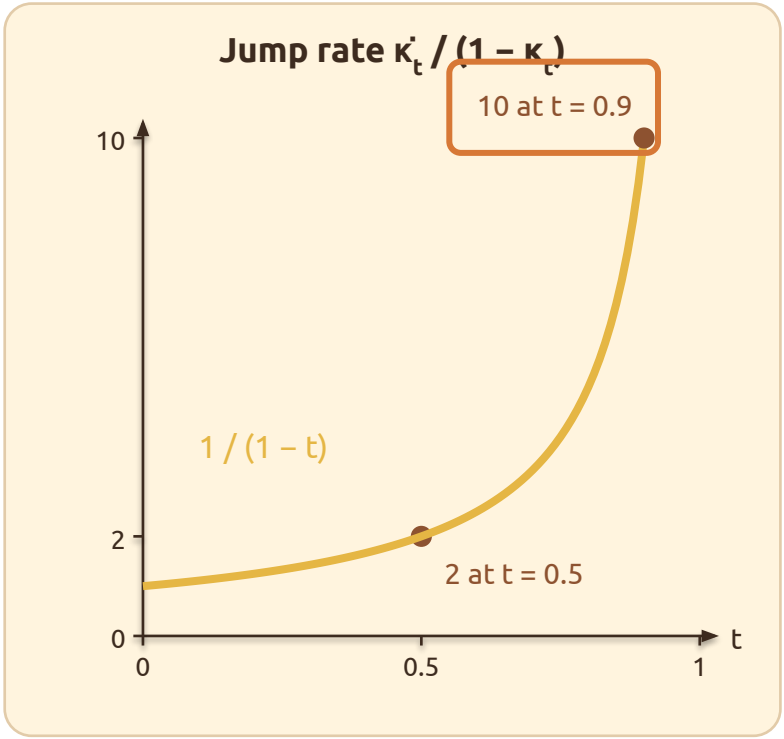
The jump rate 1/(1 – t) stays near one early and grows without bound





A late position must jump soon, so its rate is large





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In Lecture 2 and Lecture 4 we used a posterior average here, and we do the same.

Now average the conditional velocities over the pairs

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$$u_t^i(y^i, x) = \sum_{x_0, x_1} u_t^i(y^i, x \mid x_0, x_1)\, p_t(x_0, x_1 \mid x)$$

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The marginal velocity at position i, from the sequence x to the token y^i, is the sum over source and data sequences x_0 and x_1 of the conditional velocity times the posterior probability $p_t(x_0, x_1 \mid x)$ of that pair given the current sequence x.

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This theorem appears in both Discrete Flow Matching and Discrete Flow Models. It is the same statement as in Lecture 2.

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$$\boxed{p_t(x_0, x_1 \mid x) = \frac{p_t(x \mid x_0, x_1)\, \pi(x_0, x_1)}{p_t(x)}}$$

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The posterior probability of the pair (x_0, x_1) given x equals the conditional path probability $p_t(x \mid x_0, x_1)$ times the coupling $\pi(x_0, x_1)$, divided by the marginal probability $p_t(x)$.

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We can simplify this posterior to one quantity for the network to learn.

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Differentiate the marginal path, and use each conditional master equation:

$$\frac{d}{dt}\, p_t(y) = \sum_{x_0,\, x_1} \pi(x_0, x_1) \sum_{x} u_t(y, x \mid x_0, x_1)\; p_t(x \mid x_0, x_1)$$

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The rate of change of $p_t(y)$ equals the coupling-weighted sum over pairs of the conditional flux into y, $u_t(y, x \mid x_0, x_1)$ times $p_t(x \mid x_0, x_1)$, summed over x.

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Multiply and divide by $p_t(x)$, then recognize the posterior:

$$= \sum_{x} p_t(x) \Big[\sum_{x_0,\,x_1} u_t(y, x \mid x_0, x_1)\, \frac{p_t(x \mid x_0, x_1)\, \pi(x_0, x_1)}{p_t(x)} \Big] = \sum_{x} u_t(y, x)\, p_t(x)$$

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Pulling out $p_t(x)$ leaves the posterior average of the conditional velocities inside the bracket, which is the marginal velocity $u_t(y, x)$, so the marginal master equation holds.

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An average of valid velocities is a valid velocity, so the marginal velocity obeys both rules.